1. Clinical Question & Core Hypothesis

Clinical Question

Tier 1 analysis raises the question of why PI3K-targeted/metabolic therapies and genomic classifiers (PIK3CA status; proliferation-based RS) give discordant predictions of benefit — specifically, why some PIK3CA-mutant HR+/HER2– tumors do not show clinical response to PI3Kα inhibition?

Working Hypothesis

Therapy_response to PI3Kα inhibition in HR+/HER2– breast cancer is governed by a three-way interaction among tumor intrinsic PI3K activation (PIK3CA mutation), proliferation status (MKI67), and stromal content (stroma-to-tumor ratio, TSR). Specifically, PIK3CA-mutant, MKI67-high, TSR-low tumors will exhibit reduced clinical response to alpelisib (± metformin) due to tumor-cell–centric metabolic rewiring with insufficient stromal buffering to dissipate PI3K blockade–induced metabolic stress. Conversely, PIK3CA-mutant tumors with MKI67-low or TSR-high, and non-mutant tumors with favorable microenvironmental context, will show greater responsiveness. This interaction should be detectable in correlative analyses of clinical trial samples (pre-treatment MKI67 IHC, PIK3CA sequencing, TSR from pathology or radiomics, and imaging/metabolic biomarkers) and mechanistically validated in preclinical systems by modulating proliferation and stromal context in tumor organoid/co-culture models treated with PI3Kα inhibitors (with/without metformin) and assessing viability, metabolic stress markers, and compensatory pathway activation.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	PIK3CA, MKI67
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PIK3CA: PIK3CA status; proliferation-based RS) give discordant predictions of benefit — specifically, why some PIK3CA-mutant HR+/HER2– tumors do not show clinical response to PI3Kα inhibition.

MKI67: MKI67) because together they define a mechanistic axis of tumor metabolic vulnerability (PIK3CA) modulated by proliferation.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	A/B serum markers (PD/metabolic composite)	Supporting modifier from knowledge graph

Secondary	BluePrint intrinsic subtype	Supporting modifier from knowledge graph
Secondary	Oncotype DX Recurrence Score (RS)	Supporting modifier from knowledge graph
Secondary	ADC (diffusion MRI)	Supporting modifier from knowledge graph
Secondary	leptin	Supporting modifier from knowledge graph
Secondary	CEA	Supporting modifier from knowledge graph
Secondary	CA19-9	Supporting modifier from knowledge graph
Secondary	CD4 expression	Supporting modifier from knowledge graph
Secondary	KCNQ1	Supporting modifier from knowledge graph
Secondary	OSTA	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: correlative analyses within PI3Kα inhibitor trials assessing PIK3CA status, MKI67, TSR, and imaging/metabolic biomarkers against therapy_response;

Prediction 2: preclinical organoid/fibroblast co-cultures to manipulate TSR and proliferation and measure viability and metabolic stress under PI3Kα inhibition ± metformin;

Prediction 3: mechanistic readouts such as metabolic flux, lactate production, and activation of compensatory pathways (e.g., mTORC1, AMPK).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- A/B serum markers (PD/metabolic composite): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- BluePrint intrinsic subtype: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.